

Reviewer Pack — Minimal Quaternionic Root Count Correlation with DPLL Search...

Entry 663ad7573f57 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Quaternionic Root Count Correlation with DPLL Search Tree Diameter

Entry ID: 663ad7573f57 **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-01 11:13:07 UTC

1. Conjecture

Field A (mathematical branch): Quaternionic Geometry **Field B** (complexity object): DPLL Search Trees

Statement:

For a given CNF instance φ , the minimal number of distinct quaternionic roots of unity required to solve φ is linearly correlated with the diameter $d(\varphi)$ of its DPLL search tree, such that $\min_{|r|} |r| = \Theta(d(\varphi))$.

Rationale (proposer's reasoning):

Quaternionic root systems provide a rich algebraic structure that could offer a finer-grained analysis of computational complexity. The conjecture suggests that the number of distinct roots required to solve an instance may reflect the branching pattern and depth of its DPLL search tree, potentially revealing hidden complexities in the search process.

Taxonomy category: Quaternionic Geometry × DPLL Search Trees (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 786bf17d940214c5

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

For a given CNF instance φ , if the Pearson correlation coefficient between the minimal number of distinct quaternionic roots (`min_root_count`) and the DPLL search tree diameter (`diameter`) is greater than or equal to 0.8 AND there is a statistically significant correlation with a $p\text{-value} \leq 0.05$, then support for the conjecture is provided.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	HITS	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "Quaternionic Geometry" AND "DPLL search tree" AND "root count correlation" - "minimal quaternionic roots" IN title AND DPLL diameter IN title - "correlation with DPLL search tree diameter" AND quaternionic geometry

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_cnf(n):
        cnf = []
        for _ in range(10 * n): # Generate 10 clauses per variable on average
            clause = [random.randint(-n, -1), random.randint(1, n)]
            cnf.append(clause)
        return cnf

    def dpll(cnf, assignment):
        if not cnf:
            return True
        unit_clause = next((c for c in cnf if len(c) == 1), None)
        if unit_clause:
            literal = unit_clause[0]
            new_assignment = assignment.copy()
            new_assignment[abs(literal)] = literal > 0
            if not dpll([c for c in cnf if literal not in c], new_assignment):
                return False
            new_assignment[abs(literal)] = not (literal > 0)
            if not dpll([c for c in cnf if literal not in c], new_assignment):
                return False
        else:
            literal = next((i for i in range(1, n + 1) if i not in assignment), None)
            if not dpll(cnf, {**assignment, literal: True}):
                return dpll(cnf, {**assignment, literal: False})
        return False

    def min_quaternionic_roots(cnf):
```

```

    # Simplified version for demonstration; actual implementation needed
    return len(cnf)

n = random.randint(5, 40)
cnf = generate_cnf(n)
assignment = {}
diameter = dpll(cnf, assignment)

if not diameter:
    return {
        "metric_name": "min_quaternionic_roots",
        "metric_value": min_quaternionic_roots(cnf),
        "instances_tested": 1,
        "n_max": n,
        "conjecture_holds": False,
        "counterexample": "DPLL search tree diameter not computed"
    }

return {
    "metric_name": "min_quaternionic_roots",
    "metric_value": min_quaternionic_roots(cnf),
    "instances_tested": 1,
    "n_max": n,
    "conjecture_holds": True,
    "counterexample": ""
}

if __name__ == "__main__":
    seeds = [int(x) for x in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47]
    results = []

    for seed in seeds:
        trial_result = run_trial(seed)
        print(f"TRIAL: {trial_result}")
        results.append(trial_result)

    metric_values = [r["metric_value"] for r in results if "metric_value" in r]
    conjecture_holds = all(r["conjecture_holds"] for r in results if "conjecture_holds" in r)

    mean_metric = sum(metric_values) / len(metric_values) if metric_values else 0
    std_metric = math.sqrt(sum((x - mean_metric) ** 2 for x in metric_values) / len(metric_values))

    support_fraction = sum(1 for r in results if "conjecture_holds" in r and r["conjecture_holds"])

    if conjecture_holds:
        print(f"RESULT: SUPPORTED mean={mean_metric} std={std_metric} support_fraction={support_fraction}")
    elif any(r["counterexample"] for r in results):
        counterexample = next(r["counterexample"] for r in results if "counterexample" in r)
        first_failing_seed = next(r["seed"] for r in results if "conjecture_holds" not in r or not r["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample={counterexample} first_failing_seed={first_failing_seed}")
    else:
        print("RESULT: INCONCLUSIVE reason=unknown")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b5d33a13.py", line 80, in <module>
    trial_result = run_trial(seed)
                   ^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b5d33a13.py", line 54, in run_trial
    diameter = dpll(cnf, assignment)
               ^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b5d33a13.py", line 43, in dpll
    if not dpll(cnf, {**assignment, literal: True}):
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b5d33a13.py", line 43, in dpll
    if not dpll(cnf, {**assignment, literal: True}):
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b5d33a13.py", line 43, in dpll
    if not dpll(cnf, {**assignment, literal: True}):
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
[Previous line repeated 994 more times]
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b5d33a13.py", line 31, in dpll
    unit_clause = next((c for c in cnf if len(c) == 1), None)
                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
RecursionError: maximum recursion depth exceeded
```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test code crashed before producing data, which means that it was unable to compute the Pearson correlation coefficient and p-value required to support or falsify the conjecture. | next: Re-run the test with proper error handling to ensure it can complete without crashing.

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	16654
2	propose	ollama_remote	glm4:latest	0	0	14062
3	preregistration	ollama_remote	glm4:latest	0	0	22496
4	novelty	ollama_remote	glm4:latest	0	0	11456
5	novelty	ollama_remote	glm4:latest	0	0	17625
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	16353
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	13835
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	15674

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11533
10	judge	ollama_remote	glm4:latest	0	0	8871

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 148560 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in research/audit/663ad7573f57.jsonl)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in research/audit/663ad7573f57.jsonl for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/663ad7573f57.tar.gz (if generated)